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**ANOTHER NP-COMPLETE PARTITION PROBLEM  
AND THE COMPLEXITY OF COMPUTING SHORT  
VECTORS IN A LATTICE.**

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### Abstract

We prove NP-completeness for a variant of the PARTITION problem which does not require to distribute a set of numbers into two equal sum parts, but only requires to form the same sum by two different subsets. This problem is used subsequently for investigating the complexity of the computation of short vectors in a lattice in  $\mathbb{Z}^k$ .

**Keywords:** NP-completeness, Partition, Linear equation, Lattice, Euclidean and  $L_\infty$ -metric.

**AMS/MOS 1979 classification:** 68C25, 10E05.

# 1 Introduction

All numbers in this note are considered to be integral.

In this note we investigate the complexity of deciding solvability of linear equations of the form

$$a_1x_1 + a_2x_2 + \cdots + a_mx_m = b. \tag{1}$$

where the coordinates of the solution vector  $(x_1, x_2, \dots, x_m)$  are subjected to further restrictions.

For example the condition “ $x_i = 0, 1$ ” leads to the well known KNAPSACK problem ([4], problems SP13 or MP9), known to be NP-complete. Less well known is the fact that the problem remains NP-complete if the restriction on the  $x_i$  is relaxed to “ $x_i \in \mathbb{N}$ ” (Lueker [8]).

Most of the reductions used for these results are based upon the encoding of a combinatorial problem on sets like EXACT COVER ([4], problem SP2) in the bit-patterns of the integers  $a_i$ ; by “bits” in this context we understand digits equal to 1 in the radix- $d$  representation of the numbers  $a_i$ , where  $d$  is sufficiently large in order that no digit overflow will occur when performing additions. For example, in the reduction of an EXACT COVER instance with  $k$  subsets to an instance of KNAPSACK any choice of  $d$  larger than  $k + 1$  will make the reduction work.

The above technique can be adapted to prove NP-completeness in case the  $x_i$  are allowed to become negative as well, provided they remain bounded. Consider the problems defined below:

## **WEAK PARTITION**

### **INSTANCE:**

finite vector  $(a_1, a_2, \dots, a_m)$  consisting of integers.

### **QUESTION:**

is the equation  $\sum_i x_i a_i = 0$  solvable with  $x_i = -1, 0, +1$ , and not all  $x_i = 0$ ?

## **BOUNDED HOMOGENEOUS LINEAR EQUATION (BHLE)**

### **INSTANCE:**

finite vector  $(a_1, a_2, \dots, a_m)$  consisting of integers, and a positive  $K$ .

### **QUESTION:**

is the equation  $\sum_i x_i a_i = 0$  solvable with  $|x_i| \leq K$ , and not all  $x_i = 0$ ?

**THEOREM 1:** WEAK PARTITION and BOUNDED HOMOGENEOUS LINEAR EQUATION are NP-complete.

This theorem can be applied in order to obtain NP-completeness results for two more problems from the area of geometry of numbers. Consider the following problems:

**CLOSEST VECTOR**

**INSTANCE:**

a basis  $(v_1, v_2, \dots, v_k)$  for a lattice in  $\mathbb{Z}^k$ , positive  $K$ .

**QUESTION:**

does this lattice contain a vector  $x_1v_1 + \dots + x_kv_k = v$ , with  $(x_1, \dots, x_k) \neq (0, \dots, 0)$  and all coordinates of  $v$  bounded by  $K$ ?

**SHORTEST VECTOR**

**INSTANCE:**

a basis  $(v_1, v_2, \dots, v_k)$  for a lattice in  $\mathbb{Z}^k$ , positive  $K$ .

**QUESTION:**

does this lattice contain a vector  $x_1v_1 + \dots + x_kv_k = v$ , with  $(x_1, \dots, x_k) \neq (0, \dots, 0)$  and with (Euclidian) length  $|v| \leq K$ ?

We also consider an inhomogeneous variant of the latter problem.

**NEAREST VECTOR**

**INSTANCE:**

a basis  $(v_1, v_2, \dots, v_k)$  for a lattice in  $\mathbb{Z}^k$ , positive  $K$ , and a vector  $b$  in  $\mathbb{Z}^k$ .

**QUESTION:**

does this lattice contain a vector  $x_1v_1 + \dots + x_kv_k = v$ , with integral  $x_i$  such that the Euclidian length  $|v - b| \leq K$ ?

**THEOREM 2:** CLOSEST VECTOR and NEAREST VECTOR are NP-complete.

We conjecture that the same holds for the closely related SHORTEST VECTOR problem, but our proof technique fails to prove this result as well.

It will also be shown that the above three problems can be solved in polynomial time for every fixed dimension  $k$ , due to a recent algorithm for solving Integer Linear Programming problems in bounded dimension, due to H.W. Lenstra jr. [7].

## 2 Proof of Theorem 1

Starting point of our reduction is the problem PARTITION ([4], SP2):

**PARTITION**

**INSTANCE:**

a vector  $(a_1, a_2, \dots, a_m)$  consisting of integers.

**QUESTION:**

does there exist a subset  $I \subset \{1, 2, \dots, m\}$  with

$$\sum_{i \in I} a_i = \sum_{j \notin I} a_j? \quad (2)$$

Rewriting (2) in the form  $\sum_{i \in \{1, \dots, m\}} \pm a_i = 0$  one realises that the problem PARTITION is almost equal to the problem WEAK PARTITION, except for the exclusion of solution coordinates  $x_i = 0$ . In the PARTITION problem one should divide the total amount of items into two equal parts, whereas in WEAK PARTITION it suffices to produce two disjoint subsets of the items with equal sum. The impact of our result on the complexity of dividing an inheritance should by now become clear.

The “bit-pattern” engineering used in our construction can be justified by the following elementary lemma ([1, 2]):

**LEMMA:** Let the numbers  $a_i$  be expressed as  $a_i = b_i + M c_i$ , and consider the equation  $\sum_i x_i a_i = 0$ ,  $|x_i| \leq K$ . If  $M > K \sum_i |b_i|$  then this equation is equivalent to the system of equations

$$\sum_i x_i b_i = 0, \quad |x_i| \leq K,$$

$$\sum_i x_i c_i = 0.$$

**PROOF:** divide the equation  $\sum_i x_i a_i = 0$  by  $M$  and compare the integral and fractional parts, observing that the sum of the fractional parts involved is less than 1 in absolute value.

In our reduction we start with an instance of the PARTITION problem, say  $(a_1, a_2, \dots, a_m)$ . Let

$$M = 2 \sum_i a_i + 1$$

and let  $d$  be greater than 4.

For each number  $a_i$  we introduce five new numbers defined according to the scheme below:

$$\begin{aligned} b_{i1} &= a_i + M(d^{4i-4} + d^{4i-3} + 0 + d^{4i-1} + 0), \\ b_{i2} &= 0 + M(0 + d^{4i-3} + 0 + 0 + d^{4i}), \\ b_{i3} &= 0 + M(d^{4i-4} + 0 + d^{4i-2} + 0 + 0), \\ b_{i4} &= a_i + M(0 + 0 + d^{4i-2} + d^{4i-1} + d^{4i}), \\ b_{i5} &= 0 + M(0 + 0 + 0 + d^{4i-1} + 0). \end{aligned}$$

In the numbers  $b_{m2}$  and  $b_{m4}$  however the term  $Md^{4m}$  is replaced by  $M \cdot 1$ , wrapping around the  $4m$ -th bit in the  $d$ -ary representation of the integral part of the number after division by  $M$  in the 0-th position.

Repeated application of the above lemma shows that solvability of the equation

$$\sum_{i,j} x_{ij} b_{ij} = 0, \quad x_{ij} \in \{-1, 0, +1\}$$

is equivalent to solving a system of  $4m + 1$  equations, the first of which involves the original numbers  $a_i$ ; each of the other equations deals with the terms  $Md^j$  only; since for each  $j$  there exist at most 4 occurrences of such a term in the system this explains the choice  $d > 4$ .

The system of equations which results by application of the lemma can be expressed as follows:

$$\begin{aligned} 0) \quad & \sum_i (x_{i1} + x_{i4}) a_i = 0, \\ i1) \quad & x_{i1} + x_{i3} + x_{i-1,2} + x_{i-1,4} = 0, & i = 2, 3, \dots, m, \\ & x_{11} + x_{13} + x_{m2} + x_{m4} = 0, \\ i2) \quad & x_{i1} + x_{i2} = 0, & i = 1, 2, \dots, m, \\ i3) \quad & x_{i3} + x_{i4} = 0, & i = 1, 2, \dots, m, \\ i4) \quad & x_{i1} + x_{i4} + x_{i5} = 0, & i = 1, 2, \dots, m. \end{aligned} \tag{3}$$

From i2) and i3) one obtains  $x_{i2} = -x_{i1}$  and  $x_{i4} = -x_{i3}$ ; combining with i1) this yields the important fact that for every solution the quantity  $x_{i1} + x_{i3}$  does not depend on the index  $i$ ; let us call this value the *weight* of the solution. Since multiplying a solution by  $-1$  is allowed we may assume that the weight of a solution is non-negative.

Assume moreover that we are interested in solutions with  $x_{13} = 0$ ; this can be enforced by discarding the number  $b_{13}$  from our vector of numbers and considering the WEAK PARTITION instance defined by the vector of  $5m - 1$  numbers which remains. The weight of any solution now equals the value of its  $x_{11}$  coefficient. As a consequence the weight of a solution can be assumed to be either 0 or 1.

Assume that the weight of a solution equals 0. So  $x_{i1} + x_{i3} = 0$ . By i4) we have  $x_{i5} = -x_{i1} - x_{i4} = x_{i3} - x_{i1}$ ; moreover  $x_{i5} \in \{-1, 0, +1\}$ , but since  $x_{i3} - x_{i1}$  has the same parity as does  $x_{i1} + x_{i3} = 0$  we infer that  $x_{i5} = 0$ . This holds for every  $i$ , from which it is derived that our solution is the all zero solution, which is disqualified as a solution.

Similarly, in case that the weight of a solution equals 1, we know that the parity of  $x_{i5}$  is odd, so  $x_{i5} = -1$  or  $x_{i5} = +1$ ; the first case yields  $x_{i1} = 0$ ,  $x_{i3} = 1$ , whereas the other case yields  $x_{i1} = 1$ ,  $x_{i3} = 0$ .

As a consequence one obtains for each  $i$  either  $x_{i1} = 1, x_{i4} = 0$  or  $x_{i1} = 0, x_{i4} = -1$ .

Next reconsider equation 0); what we observed above about  $x_{i1}$  and  $x_{i4}$  shows that the sum  $x_{i1} + x_{i4} = \pm 1$ . So the solvability of 0) implies the fact that the given vector  $(a_1, \dots, a_m)$  allows a solution of the partition problem.

Conversely, it is easy to see that, given a positive solution for the PARTITION problem for the vector  $(a_1, \dots, a_m)$  a corresponding solution for the WEAK PARTITION problem for the vector

$$(b_{11}, b_{12}, b_{14}, b_{15}, b_{21}, \dots, b_{m5})$$

is obtained. Since the transformation which translates the given vector  $(a_1, \dots, a_m)$  into this vector of  $5m - 1$   $b$ 's clearly is polynomial this proves the NP-completeness of WEAK PARTITION.

For the NP-completeness of BOUNDED HOMOGENEOUS LINEAR EQUATION we see a variant of the above reduction. First replace the number  $d$  by  $Kd$ , in order to preserve validity of the application of the lemma. Next we replace all terms of the form  $Md^{4i-j}$  by  $KMd^{4i-j}$ , except for the numbers  $b_{i5}$  which remain to be defined by  $Md^{4i-1}$ . This leaves the system of equations (3) unchanged, except for the equations i4) which are replaced by:

$$\text{i4')} \quad K(x_{i1} + x_{i4}) + x_{i5} = 0.$$

Since  $|x_{i5}|$  is bounded by  $K$  this enforces  $|x_{i1} + x_{i4}|$  to be bounded by 1. In particular, since  $x_{i4} = 0$ , the weight of a solution is  $-1, 0$  or  $1$ ; the case  $-1$  reduces to weight  $+1$  by multiplying the solution with  $-1$ .

Again we want to show that the only solution with weight 0 is the trivial all zero solution; this is indeed a consequence of the equation  $x_{i1} + x_{i3} = 0$  combined with the inequality  $|x_{i1} - x_{i3}| \leq 1$ . Finally for a solution with weight 1 we conclude  $x_{i1} + x_{i4} = -1$  or  $+1$ , providing us in 0) with a solution of the PARTITION problem for the vector  $(a_1, \dots, a_m)$ .

It should be remarked at this point that the instances of WEAK PARTITION and BHLE which are obtained by our reduction have the property that just a doubling of the bound 1, or  $K$  respectively, suffices to obtain a large collection of "parasitic weight zero solutions": for every  $j \geq 2$  one has that

$$b_{j1} + b_{j4} = b_{j2} + b_{j3} + 2b_{j5} + 2a_j,$$

or

$$b_{j1} + b_{j4} = b_{j2} + b_{j3} + 2Kb_{j5} + 2a_j,$$

respectively. These solutions cause the failure of our method to prove NP-completeness for the SHORTEST VECTOR problem.

### 3 The Complexity of Computing Shortest Vectors; Proof of Theorem 2

The problem of computing a shortest non zero vector in a lattice has drawn the attention from Number Theoreticians, see EG. [3]. In the two-dimensional case there exists a reduction algorithm due to Gauss [5] which transforms a given pair of two independent vectors  $a_1$  and  $a_2$ , into a “reduced” pair  $b_1$  and  $b_2$ , such that: 1)  $\mathbb{Z}a_1 + \mathbb{Z}a_2 = \mathbb{Z}b_1 + \mathbb{Z}b_2$ , and 2)  $b_1$  is a shortest non-zero vector in the lattice spanned by  $a_1$  and  $a_2$  and  $b_2$  is a shortest vector in this lattice independent of  $b_1$ . Lagarias [6] has established that this algorithm runs in polynomial time.

In this section we draw attention to a negative and a positive result. The negative result is given by Theorem 2 which indicates that for the  $L^\infty$ -metric the problem of computing a shortest non-zero vector in a lattice is NP-hard. The positive result is that for every fixed dimension  $k$  the problem of finding a shortest vector in a  $k$ -dimensional lattice is polynomial solvable, due to a recent algorithm for solving integer linear programming problems in fixed dimension described by H.W. Lenstra jr. [7].

In the formulation of the SHORTEST VECTOR and CLOSEST VECTOR problem we have tacitly assumed that the number of vectors in the set  $(v_1, \dots, v_k)$  equals the dimension of the space  $\mathbb{Z}^k$ . This is not a restriction of the generality of the problem. First consider the case that the vectors  $v_1, \dots, v_k$  are not linearly independent, which fact can be verified in polynomial time; in this case the zero vector can be written as the sum of a non trivial integral combination of the given vectors and the algorithm simply answers “yes”. This shows that all instances of a generalisation with more vectors than the dimension of the space are trivial. A more interesting generalisation consists of the case where the number of vectors  $k'$  is less than the dimension of the space  $k$ , and where the  $k'$  vectors are independent. Such an instance can be reduced to an instance of the problems as expressed by extending the list to a complete basis with a set of  $k - k'$  vectors which will never occur in a linear combination forming a short vector; for example a set of  $k - k'$  mutually orthogonal vectors, which are moreover orthogonal to the subspace spanned by the given vectors, and whose length is greater than  $kK$  will be adequate. It is easy to see that such an extension can be computed in polynomial time.

NP-completeness for the CLOSEST VECTOR problem is obtained by a reduction showing that this problem is a rather direct translation of the BOUNDED HOMOGENEOUS LINEAR EQUATION problem.

Let  $((b_1, \dots, b_m), K)$  denote an instance of BHLE. Consider the following set of vectors  $v_j$  in  $\mathbb{Z}^{m+1}$ ,  $j = 1, \dots, m+1$ . We denote the coefficients of  $v_j$  by  $v_{j1}, \dots, v_{j,m+1}$ :

$$\begin{aligned} v_{j,i} &= 0 & \text{for } i \neq j, \quad 1 \leq i, j \leq m, \\ v_{j,j} &= 1 & \text{for } 1 \leq j \leq m, \\ v_{j,m+1} &= K' b_j & \text{for } 1 \leq j \leq m, \quad \text{where } K' = K + 1, \\ v_{m+1,i} &= 0 & \text{for } 1 \leq i \leq m, \\ v_{m+1,m+1} &= K'' & \text{where } K'' = 2K' \sum_j b_j + 1. \end{aligned}$$

Suppose that  $x_1 v_1 + \dots + x_m v_m + x_{m+1} v_{m+1} = w$  is a non trivial integral combination of the vectors  $v_j$  such that all coordinates of  $w$  are bounded by  $K$ . Denote the coordinates of  $w$  by  $w_1, \dots, w_m, w_{m+1}$ . Now it follows that  $w_j = x_j$  for  $1 \leq j \leq m$ , so  $x_j$  is bounded by  $K$ . Moreover for  $w_{m+1}$  we have the equality

$$w_{m+1} = (x_1 b_1 + \dots + x_m b_m) K' + x_{m+1} K'.$$

Due to the relative size of  $K''$  and  $K'$  we infer that  $x_{m+1} = 0$  and  $x_1 b_1 + \dots + x_m b_m = 0$ . This shows that the instance  $((v_1, \dots, v_m, v_{m+1}), K)$  of CLOSEST VECTOR is solvable if and only if the given instance of BHLE was solvable. So CLOSEST VECTOR is NP-hard. That CLOSEST VECTOR belongs to the class NP can be shown by the same method as used for the general Integer Linear Programming problem.

That a reduction like the one above does not work for the case of the Euclidean metric (SHORTEST VECTOR) can be seen as follows. Assuming that we start with the same instance  $((b_1, \dots, b_m), K)$  of BHLE, and assume moreover that this instance is solvable; so there exists a non trivial combination  $x_1 b_1 + \dots + x_m b_m = 0$ . Under the above reduction this would yield a vector in the lattice with Euclidean length

$$(x_1^2 + \dots + x_m^2)^{1/2} \leq m^{1/2} K.$$

If we consider however the instances of BHLE which were produced in order to prove NP-hardness of BHLE we infer that at least  $m/5$  of the  $x_i$  are  $\pm K$  in any positive solution of these instances; so in order not to forsake the solutions which will make the reduction work the bound on the vector length should not become less than  $5^{-1/2} m^{1/2} K$ . But as soon as  $m > 20$  this opens the possibility for individual coefficients  $x_j$  to become as large as  $2K$ , and therefore the “parasitic” weight zero solutions mentioned at the end of the preceding section will come in existence.

The above difficulty can be evaded by considering an inhomogeneous variant of the SHORTEST VECTOR problem which we have introduced as the NEAREST VECTOR problem.

Start again with an instance of PARTITION:  $(a_1, \dots, a_{m'})$  and transform it into an instance of WEAK PARTITION  $(b_1, \dots, b_m)$  where  $m = 5m' - 1$ , taking  $d > 2m'$ . If this instance has a solution at all, then we may assume for this solution

$$x_1 b_1 + \dots + x_m b_m = 0, \quad |x_i| \leq 1,$$

that the weight of this solution is not zero and equal to  $x_1$ . Moreover the number of non zero  $x_j$  equals  $3m'$ .

Consider the collection of vectors  $v_1, \dots, v_{m+1}$  constructed as in the previous reduction, taking  $K = m + 1$ . Let  $u_1$  be the vector with coordinates  $u_{11}, \dots, u_{1,m+1}$ , where  $u_{11} = K''$ , and  $u_{1i} = 0$  for  $2 \leq i \leq m + 1$ . Finally consider the instance of the NEAREST VECTOR problem defined by

$$((u_1, v_2, \dots, v_{m+1}), (3m')^{1/2}, v_1).$$

If the original instance of PARTITION is solvable, then the WEAK PARTITION instance will have a solution with  $x_1 = -1$ . Writing this solution in the form

$$b_1 = x_2 b_2 + \dots + x_m b_m$$

with exactly  $3m' - 1$  of the  $x_j \neq 0$ , we infer that the vector  $w = x_2 v_2 + \dots + x_m v_m$  satisfies that the Euclidean distance between  $v_1$  and  $w$  is equal to  $(3m')^{1/2}$ .

Conversely, assume that  $w = x_1 v_1 + \dots + x_{m+1} v_{m+1}$  satisfies that the Euclidean norm  $|v_1 - w|$  is bounded by  $(3m')^{1/2}$ . Due to the size of  $K''$  we infer that  $x_1 = x_{m+1} = 0$ . It also follows that

$$b_1 = x_2 b_2 + \dots + x_m b_m,$$

whereas the  $x_i$  are bounded by

$$(3m')^{1/2} < m < \frac{1}{2}d.$$

In our analysis of this solution we can apply therefore the Lemma from section 2 and infer that we have a proper solution with weight  $-1$ ; hence this solution originates from a positive solution of the original PARTITION instance.

It follows that we have reduced the PARTITION problem to the NEAREST VECTOR PROBLEM, using WEAK PARTITION as an intermediate step. So NEAREST VECTOR is NP-hard; that this problem belongs to the class NP is shown by the same methods as used for the general Integer Linear Programming problem.

Finally we consider the complexity of solving the CLOSEST VECTOR and SHORTEST VECTOR problem for bounded dimension  $k$ . Recall from [7] that there exists for each  $k$  a polynomial algorithm which, given a system of inequalities

$$\begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nk} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix} \leq \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}, \quad (4)$$

decides whether this system has a solution with integral  $x_i$ ; if such a solution exists, it will moreover produce an explicit solution.

Let there be given a collection of vectors  $v_1, \dots, v_k$  in  $\mathbb{Z}^k$  which is a basis for some lattice. The question whether there exists a non-zero vector in this lattice with all coordinates bounded by  $K$  can be decided by solving the following  $K$  Integer Linear Programming problems. Let  $v_j$  be the vector  $(v_{j1}, \dots, v_{jk})$  and consider the  $2k$  by  $k$  matrix  $A$  defined by

$$A = \begin{pmatrix} v_{11} & \cdots & v_{1k} \\ -v_{11} & \cdots & -v_{1k} \\ \vdots & & \vdots \\ v_{k1} & \cdots & v_{kk} \\ -v_{k1} & \cdots & -v_{kk} \end{pmatrix}$$

and consider the vectors  $b_j$  defined by

$$b_j = (b_{j1}, b_{j2}, \dots, b_{j,2k}), \quad j = 1, \dots, k,$$

with

$$b_{j,2i-1} = K, \quad b_{j,2j} = -1, \quad b_{j,2i} = -K \quad \text{for } i \neq j.$$

A solution of  $Ax \leq b_j$  corresponds to a linear combination of the vectors  $v_j$  whose sum has coordinates bounded by  $K$  and positive  $j$ -th coordinate; if any of these Integer Linear Programming problems has a solution then the CLOSEST VECTOR problem has a positive solution, and conversely, if there exists a short non zero vector in the lattice it may be assumed to have a positive coordinate. Note that directly asking for the existence of a lattice point in the hypercube of radius  $K$  centered around the origin is not a proper translation of the question since the origin itself always belongs to the lattice.

A closest vector now can be determined by computing, using binary search, the smallest radius of a cube containing a non-zero lattice point.

For the SHORTEST VECTOR problem one might attempt a similar covering of the set of non-zero points close to the origin in  $\mathbb{Z}^k$  by a finite collection of half spheres of radius  $K$ , but a description of these convex sets by linear inequalities may become rather large. Instead we again apply the relation between the Euclidean and the  $L^\infty$ -metric:

$$|v|_\infty < |v|_2 < k^{1/2}|v|_\infty.$$

First try to find a non-zero vector in the lattice with all coordinates bounded by  $k^{-1/2}K$ ; if there exists such a vector this vector is located in the sphere with radius  $K$ . If not then we know that each hypercube with diameter  $k^{-1/2}K$  contains at most a single lattice point. The sphere with radius  $K$  can be covered by no more than  $(2k^{1/2} + 1)^k$  hypercubes of diameter  $k^{-1/2}K$ ; solving each of these Linear Integer Programming problems will produce a complete enumeration of all lattice points within the sphere, and a few more close to the surface of the sphere; the shortest of these vectors can be determined by exhaustive search.

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## Note Added in Print

The result that WEAK PARTITION is NP-complete has been proved independently by Arthur L. Rubin (abstracts paper presented at AMS meetings 783–10–50, Jan 1981).